

Summary for 2024-11-07-NetSimplex

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1 Summary

This lecture introduced the Network Simplex Method, a specialized and efficient approach within linear programming for solving network optimization problems. The method leverages the structure of networks and spanning trees to find optimal solutions.

Network: A network is defined as a pair (N, A) , where N represents the nodes and A represents the arcs. **Spanning Tree:** A spanning tree $T = (N, A)$ is a connected, acyclic subgraph of a network, easily found using standard algorithms. **Primal Flow:** A primal feasible solution (flow) always exists and can be found by starting from a leaf node and moving inwards, though it might initially contain negative flow values. **Dual Flow:** A dual feasible solution (flow) always exists, defined by equations relating dual variables at nodes, reduced costs on arcs, and arc costs. Negative reduced costs indicate non-optimality. **Spanning Tree Properties:** Adding one arc to a spanning tree creates a cycle; removing one arc disconnects the tree. These properties are fundamental to the algorithm's efficiency. **Primal Network Simplex Method:** This iterative method involves selecting an entering arc (with negative reduced cost), creating a cycle, selecting a leaving arc from the cycle, and updating primal and dual flows to reduce the cost. The leaving arc is chosen based on its orientation and minimum flow within the cycle. **Dual Network Simplex Method:** This method starts with a dual feasible but potentially primal infeasible solution (negative flows). It iteratively improves the solution by selecting a leaving arc (with negative flow) and an entering arc (in the opposite direction with the smallest dual slack) to maintain dual feasibility and move towards primal feasibility. **Algorithm Variants:** Multiple variants of the Network Simplex Method exist, including two-phased approaches (using dual then primal simplex) and a parametric self-dual method, mirroring similar techniques in standard linear programming. These variants often involve temporary modifications to costs or supplies to ensure initial feasibility.